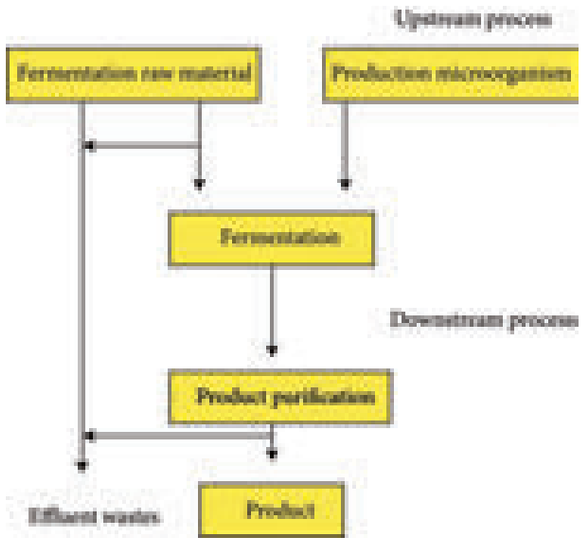


Figure 6.7: An overview of Fermentation process

Inoculum development

It is a preparation of a population of micro organisms from a stock dormant culture to a state useful for inoculating a final production fermentor.

It is a critical stage in fermentation process.

It is a stepwise sequence employing increasing volume of media.

Inoculum media usually balanced for rapid cell growth and not for product formation.

Inoculum scale up

It is the preparations of the seed culture in amounts sufficient to be used in the larger fermenter vessel. It involves growing the microorganisms obtained from the pure stock culture in several consecutive fermenter. By doing this, the time required for the growth of microbes in the fermenter is cut down, so that the rate of productivity is increased. The seed culture obtained is then used for inoculation in fermentation medium. The size of the inoculums is generally 1–10% of the total volume of the medium.

In general, fermentation/ bioprocess techniques are developed in stages starting from a laboratory and finally leading to an industry. The phenomenon of developing industrial fermentation process in stages is referred to as scale–up. Scale–up is necessary for implementing new fermentation technique developed using mutant organisms.

The very purpose of scale–up is to develop optimal environmental and operating conditions at different levels for a successful fermentation industry where conditions like substrate concentration agitation and mixing, aeration, power consumption and rate of Oxygen transfer are studied. In a conventional scale–up, a fermentation technique is developed in 3–4 stages. The initial stage involves a screening process using Petri dishes or Erlenmeyer flasks followed by a pilot project to determine the optimal operating conditions for a fermentation process with a capacity of 5–200 litres. The final stage involves the transfer of technology developed in the laboratory to industry.(Figure 6.8)

It has to be continuously noted that a fermentation process that works well at the

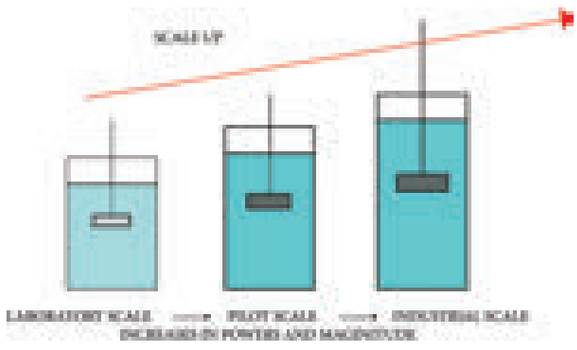


Figure 6.8: Scale up in industrial fermentation

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